

Attribution without a handle, removal without removal: a pedagogical tour of feature-resolved attention on toy HMMs

FRA theory sprint

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TL;DR. On a toy hidden-Markov task where every quantity has a closed form, attention-pattern edits guided by feature-level attribution turn out to be causally empty — the feature×feature terms carry 57–75% of the attention-score mass, yet cutting them removes at most 10% of the concept’s behavioral effect — while a gain-tuned edit to the attention *output* erases the behavior completely and yet a freshly retrained probe still reads the concept from the activations at $R^2 = 0.985$. This note explains both surprises from first principles. The tools are elementary: one identity about softmax rows, one observation about running averages, and one bookkeeping freedom in how a residual stream splits into “token content” and “positional content.” Every formal statement here is proved in the companion theory note (`fra_theory_note.tex`); this note builds each one up from concrete numbers first. No prior exposure to the theory note, or to the underlying papers, is assumed.

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1 The puzzle: four interventions, one scoreboard

1.1 The experiment, from zero

The experiment (`experiments/fra_hmm_toy`) was built to answer one question with no unknowns left in it: *when you try to remove an in-context concept from a small transformer, what do the different surgical tools actually do?*

On real models this question is confounded — maybe the sparse autoencoder is bad, maybe the concept was never found, maybe the edit misses its target. The toy removes each confound by making every target analytically known. **The data.** Three small hidden-Markov components (from the Mess3 family, each with 3 hidden states and 3 symbols) run in parallel, at three dynamical regimes: component 0 is sticky ($x = 0.08$, $a = 0.90$; long same-symbol runs), component 1 is moderate ($x = 0.25$, $a = 0.65$), component 2 is diffuse ($x = 0.40$, $a = 0.34$; near-memoryless). Each sequence first draws a hidden mixing vector

$$\omega \sim \text{Dirichlet}(10 \cdot [0.40, 0.35, 0.25]),$$

then at every timestep picks a component $c_t \sim \text{Cat}(\omega)$ and lets that component emit one symbol from its current hidden state. The three components write into *disjoint vocabulary blocks*: component c emitting symbol s produces the global token $v = 3c + s$, so tokens $\{0, 1, 2\}$ belong to component 0, $\{3, 4, 5\}$ to component 1, $\{6, 7, 8\}$ to component 2. Every token therefore announces *which* component produced it (its *block tag*, $\lfloor v/3 \rfloor$). What no token announces is ω itself.

The concept is literally a count. The per-sequence mixing vector ω is the “concept” whose removal we study. Its Bayes-optimal estimate after $t + 1$ tokens, with per-block counts $n = (n_0, n_1, n_2)$, is the Dirichlet posterior mean

$$\hat{\omega}(t) = \frac{\alpha + n}{10 + t + 1}, \quad \alpha = 10 \cdot (0.40, 0.35, 0.25) = (4, 3.5, 2.5). \quad (1)$$

You can check this by hand on the sequence used throughout the toy’s write-up (eval sequence #270, true $\omega = (0.139, 0.834, 0.027)$): after 17 tokens the block counts are $n = (2, 13, 2)$, so $\hat{\omega} = (4+2, 3.5+13, 2.5+2)/27 = (0.222, 0.611, 0.167)$, matching the pipeline’s stored posterior digit for digit. Keep Eq. (1) in view: *the concept is a running average of block tags*, and Section 4 will show that this single structural fact protects it from an entire family of interventions.

The loss splits the concept off exactly. The Bayes-optimal next-token prediction factorizes: $p^*(v = 3c + s \mid \text{context}) = \hat{\omega}_c(t) \cdot p_c(s \mid \text{belief}_c(t))$ — a *concept* factor (which block comes next) times a *grammar* factor (which symbol within the block, governed by the component’s belief-state dynamics). The log-loss therefore splits additively into a block cross-entropy and a within-block cross-entropy, with analytic anchors on the eval set (500 sequences $\times$ 256 positions, $t \geq 8$): Bayes block CE 0.9872 nats, no-concept (prior) block CE 1.0809, so *the concept is worth 0.0937 nats per token*; Bayes within-block CE 0.7785, uniform $\log 3 = 1.0986$, so the grammar range is 0.3201 nats. Every removal metric lives in the first term, every collateral metric in the second, with no leakage by construction.

The model and the SAE. A 3-layer transformer ($d_{\text{model}} = 64$, 2 heads, trained on random crops of long training sequences) learns the task nearly perfectly: its block CE is 0.9881 — it uses 99% of the available concept information — and its *block-mass vector* $m(t)$ (the probability it assigns to each vocab block for the next token, i.e. its operational concept estimate) tracks the Bayes posterior $\hat{\omega}(t)$ with $R^2 = 0.992$. A TopK SAE (64 latents, $k = 4$ active per position) is trained on the residual stream; with TopK, the residual at each position decomposes *exactly* into active-latent contributions plus a decoder bias plus a reconstruction error, and the toy verifies the resulting attention-score decomposition against the model’s own scores to $\leq 3 \times 10^{-5}$. The cut set S used everywhere is **omega_top4**: the latents that best track the posterior $\hat{\omega}$ (selected on training data; all metrics on held-out data).

The three interventions, all forward hooks, all with gain parameters:

- **SAE cut** (content removal at the source): subtract $\alpha \sum_{j \in S} z_j d_j$ from the residual wherever the latents fire. At $\alpha = 1$ the latents’ content is gone.
- **FRA-QK cut** (routing removal): the attention scores of the downstream layer decompose exactly into term-pair contributions (query-side term $\times$ key-side term, each side one of {latents, decoder bias, error, constant}). Subtract c times the summed contribution of every pair involving S . At $c = 1$, S ’s influence on *who attends to whom* is exactly gone.
- **FRA-OV cut** (path-restricted content removal): subtract c times the content of S that this attention layer *transports*, computed with the clean pattern. At $c = 1$ this severs exactly this path’s share of S -content; at $c > 1$ it is no longer severing — it injects $-(c - 1) \times$ the contribution as a counter-signal. This distinction becomes the whole story of Section 5.

The metrics, all normalized by the analytic anchors:

- **RF** (removal fraction): how much of the 0.0937-nat concept value is destroyed. 0 = untouched; 1 = degraded exactly to the no-concept prior.
- **CF** (collateral fraction): how much of the 0.3201-nat grammar range is destroyed.

- **tracking** R^2 : between the model’s block-mass $m(t)$ and the Bayes posterior $\hat{\omega}(t)$; clean 0.992, perfect removal ≈ 0 .
- **presence**: R^2 of a ridge probe *retrained from scratch* on the intervened activations, predicting $\hat{\omega}$. This is the adversarial “is the concept still decodable” metric — it refits, so moving the representation around cannot fool it. Clean: 0.989 (linear), 0.996 (MLP).

1.2 The scoreboard

intervention (best per family)	RF (use removed)	CF (collateral)	presence
SAE cut, ω -latents, $\alpha = 1$	0.86	0.017	0.49
FRA-QK (every set, side, gain, layer)	≤ 0.10	up to 0.30	0.99
FRA-OV, $c = 1$ (exact severing)	0.06	0.002	0.99
FRA-OV, $c \approx 4$ (gain-tuned)	0.99	0.036	0.985
random-latent controls	0.00	0.00	0.99

Two rows should bother you.

Puzzle 1: large attribution, zero handle. At the intervened interfaces, feature×feature term-pairs carry 57–75% of the pattern-weighted attention-score mass. By the usual attribution logic, the features dominate the routing, so cutting their routing contribution should matter. It removes at most 10% of the concept’s use — at every feature set, every side (key/query/either), every gain, every layer — while collateral grammar damage climbs to 0.30. The attribution found real mass and the mass was causally empty.

Puzzle 2: full behavioral removal, zero representational removal. Severing the OV path exactly ($c = 1$) removes almost nothing (RF 0.06). Overdriving the same edit to $c \approx 4$ produces a full behavioral null: RF 0.99, tracking $R^2 = 0.001$, collateral only 0.036. The model passes any on-distribution behavioral test of “does it use ω .” And yet a probe retrained on the intervened activations reads $\hat{\omega}$ at $R^2 = 0.985$ (linear) / 0.995 (MLP), versus clean 0.989/0.996. The concept is computed, represented, intact — and unexpressed.

The rest of this note derives both puzzles from three elementary facts, in increasing order of specificity: what a softmax row is (Section 2), what an attribution split can and cannot pin down (Section 3), and what a running average does to key-to-key differences (Section 4). Section 5 then reads the OV null quantitatively — including a prediction of the full intervention sweep from the *clean* model alone — and Section 6 shows that one dial, the spectral gap of the data process, interpolates between the regime where FRA works and the regime studied here. Section 7 lists what remains unproven or imperfect.

2 One softmax row is a probability distribution

Everything about pattern edits follows from taking one sentence seriously: *each row of an attention pattern is a probability distribution over keys, and the head’s output is an expectation under it.*

Fix a query position d and one head. The clean scores over keys $s \leq d$ are σ_s , the pattern is $A_s = e^{\sigma_s} / \sum_{s'} e^{\sigma_{s'}}$, each key contributes a value vector v_s (its content, mapped through the OV matrices), and the head’s output at d is

$$c_d = \sum_{s \leq d} A_s v_s = \mathbb{E}_A[v].$$

Every FRA-QK cut — indeed every score edit whatsoever — has the form $\sigma \rightarrow \sigma - c \delta$ for some per-key numbers δ_s and gain c . What happens to the row?

The tilt identity. The edited row is an exponential reweighting of the clean row,

$$A_s^c = \frac{A_s e^{-c \delta_s}}{\mathbb{E}_A[e^{-c \delta}]}, \quad (2)$$

and the output change is exactly a covariance: $\Delta c_d = \text{Cov}_A(e^{-c \delta}, v) / \mathbb{E}_A[e^{-c \delta}]$. To first order in c ,

$$\Delta c_d = -c \text{Cov}_A(\delta, v) + R, \quad \|R\| \leq \frac{c^2}{8} \text{osc}(\delta)^2 D_v, \quad (3)$$

where $\text{osc}(\delta) = \max_s \delta_s - \min_s \delta_s$ is the spread of the edit across the row and D_v is the diameter of the value set. The remainder bound is unconditional — any edit, any weights, any data.

Read Eq. (3) as a slogan: **a score edit moves the output only insofar as the edit covaries with the values across the row.** An edit that is large but flat across keys does nothing; an edit that singles out keys whose values agree with the row average does almost nothing. Both halves of that slogan are exact statements, and both will fire in Sections 3 and 4.

2.1 A worked example with three keys

(The numbers in this subsection are a made-up illustration, chosen for round arithmetic; every other number in this note comes from the experiments.)

Take a row with three keys, clean scores $\sigma = (1, 2, 3)$, so

$$A = \frac{(e^1, e^2, e^3)}{e^1 + e^2 + e^3} = (0.0900, 0.2447, 0.6652),$$

and scalar values $v = (1, 4, 2)$. The clean output is $\mathbb{E}_A[v] = 0.0900 + 4(0.2447) + 2(0.6652) = 2.3994$.

Now cut key 2: $\delta = (0, 1, 0)$, i.e. subtract c from the second score. The tilt identity multiplies key 2’s unnormalized weight by e^{-c} and renormalizes:

	A_1^c	A_2^c	A_3^c	output	Δ
clean ($c = 0$)	0.0900	0.2447	0.6652	2.3994	—
$c = 0.5$	0.0996	0.1643	0.7361	2.2289	−0.1705
$c = 1$	0.1065	0.1065	0.7870	2.1065	−0.2929

Three things to notice, each of which scales up to the real experiment:

1. **The removed mass goes to the survivors in proportion to their clean weights.** At $c = 1$, keys 1 and 3 each grow by the same factor 1.183. Softmax renormalization is automatic redistribution; a “cut” is a reallocation.
2. **The first-order formula is good, and its error is exactly quadratic.** Here $\text{Cov}_A(\delta, v) = \mathbb{E}_A[\delta v] - \mathbb{E}_A[\delta] \mathbb{E}_A[v] = 0.9789 - 0.2447 \times 2.3994 = 0.3917$. The first-order prediction $-c \times 0.3917$ gives -0.1959 at $c = 0.5$ (exact: -0.1705 , relative error 15%) and -0.3917 at $c = 1$ (exact: -0.2929 , relative error 34%). The remainder bound $\frac{c^2}{8} \text{osc}(\delta)^2 D_v = \frac{c^2}{8} (1)^2 (3)$ gives 0.094 and 0.375; both errors (0.025 and 0.099) sit comfortably inside. Doubling c quadrupled the absolute error ($0.025 \rightarrow 0.099$), as a c^2 remainder must; the relative error doubles, since the exact response itself grows roughly linearly.
3. **Edits of this shape have an exact closed form at every gain.** For a cut $\delta = \kappa \mathbb{I}[s \in S]$ with clean mass w on S : $\Delta c_d = (e^{-c\kappa} - 1) w (\mu_S - \bar{v}) / (1 + (e^{-c\kappa} - 1)w)$, where μ_S is the mean value on the cut keys and $\bar{v}$ the row mean. Plugging in $w = 0.2447$, $\mu_S = 4$, $\bar{v} = 2.3994$, $\kappa = 1$, $c = 1$ reproduces -0.2929 to four digits. As $c \rightarrow \infty$ this saturates at “delete the S keys and renormalize” — a score cut can never do more than that.

The same calculus was tested on the real toy model (checkpoint `phase2.L0`, key-side `omega_top4` cuts): across every head-row at gains $c = 0.5, 1, 2$, the remainder bound of Eq. (3) holds with *zero violations*, and the median first-order relative error grows $15\% \rightarrow 31\% \rightarrow 61\%$ — the same quadratic staircase as the three-key example. The calculus is exact as stated, on trained weights.

2.2 The two invariances

Two special cases of the slogan deserve their own box, because between them they explain almost everything in the scoreboard.

Invariance 1 (flat edits are inert). If δ_s is the same number for every key in the row — a *per-row constant* — then $e^{-c\delta_s}$ cancels between numerator and denominator of Eq. (2): $A^c = A$ exactly, at every gain. Softmax only sees score *differences* within a row.

Invariance 2 (agreeing keys are immune). If a readout u satisfies $u \cdot v_s = h_0$ for every key in the row, then under *any* row-stochastic pattern whatsoever — any edit, any gain, including edits far outside the FRA family — $u \cdot c_d = h_0 \sum_s A'_s = h_0$. A readout on which all keys agree is beyond the reach of every pattern intervention.

Invariance 1 is a statement about the *edit*; Invariance 2 is a statement about the *readout*. Section 3 shows that at the trained optimum for this kind of data, FRA-QK edits are per-row constants plus gauge — Invariance 1 territory. Section 4 shows that the mixture concept makes all keys *nearly* agree — approximate Invariance 2 territory, with the approximation error controlled by an explicit drift bound. The two mechanisms are logically independent, and the toy sits in the intersection.

3 Puzzle 1, part one: the feature×feature mass is a bookkeeping coordinate

3.1 What the optimal pattern looks like for this data

For a stationary hidden-Markov process, the best next-token predictor available to a one-layer attention model is a *constrained belief update*: the belief displacement after context $z_1 \cdots z_d$ is

$$r_1(d) = \pi + \sum_{s \leq d} \lambda^{d-s} g(z_s) \quad (\text{one term per eigenvalue } \lambda \text{ of the hidden chain}), \quad (4)$$

where $g(z)$ is the fixed belief displacement contributed by token z and λ^{d-s} discounts it by age. Look at the two factors. The discount λ^{d-s} depends only on the *lag* $d - s$. The content $g(z_s)$ depends only on the *token*. A single attention head computes exactly such a sum when its pattern supplies the lag weights and its OV matrices supply g : the pattern’s optimal shape is *lag-only* — the same for every sequence, blind to what the tokens are. All token information travels through the values. (For softmax cross-entropy models this lag-only optimality is a conjecture, supported by trained-model measurements and proved for an MSE surrogate; Section 7. Everything below is exact *given* a token-independent pattern.)

Here is the tension. The trained model realizes a token-independent pattern, so by construction the token identities have no influence on who attends to whom. Yet FRA-QK attribution on such a model reports that feature×feature terms — token content on the query side times token content on the key side — carry 57–75% of the score mass. Both facts are true simultaneously. The resolution is that the attribution is measuring a coordinate that the model’s behavior never pinned down.

3.2 The re-split: same model, any attribution

The residual stream at position s is one vector: $\mathbf{x}_s = e(z_s) + p_s$, token embedding plus positional embedding. The split into “token part” and “position part” is bookkeeping — the forward pass only ever consumes the sum. So consider the reparameterization

$$e(z) \rightarrow e(z) + u, \quad p_s \rightarrow p_s - u, \quad \text{for any fixed vector } u.$$

Every residual $\mathbf{x}_s$ is *bit-identical*. Every activation, the loss, and the behavior on and off distribution are bit-identical. But the FRA blocks are defined in terms of the split: the feature×feature score contribution $q_E(z_d) \cdot k_E(z_s)$ involves $e(z)$ on both sides, and adding u to every token embedding moves it. Dialing u dials the feature×feature mass from zero to arbitrarily large — at literally unchanged weights-as-a-function.

This is one member of a family of loss-invariant reparameterizations (the companion note’s gauge families G1–G3; the others live in the $\mathbf{W}_Q/\mathbf{W}_K$ weights and are exact on-distribution). The theorem they add up to:

The QK gauge theorem (informal). At any configuration whose pattern is token-independent, the entire token-dependent structure of the feature×feature score mass consists of *gauge coordinates*: quantities that loss-preserving reparameterizations dial from exactly 0 to arbitrarily large, with identical behavior everywhere. The same holds for every FRA-QK cut coefficient. FRA-QK attribution at such an optimum measures which gauge training happened to select, and the softmax then deletes the selected gauge: the varying pieces enter every row as per-row constants or as cancelling cross-terms, so Invariance 1 applies.

The analogy that carries the exact structure is longitude: the prime meridian is a real, measurable coordinate on every map, and moving it changes every longitude value while changing no distance between any two cities. The 57–75% figure is a longitude. The loss level set provably contains configurations where that number is 0% and configurations where it dominates, all computing the same function. Interpreting the trained value causally is the error the theorem licenses you to name.

3.3 The gauge dial, performed on a real model

The sprint’s Phase A trains exactly the model this theorem talks about: one layer, one head, on single Mess3 (no mixture), reaching CE 1.08726 against a Bayes optimum of 1.08656 — 94% of the available in-context information. Its trained pattern is token-independent to a measured 9.4% relative spread (max 19.6%): that spread is the content leak ε , the model’s actual distance from the theorem’s idealization. Two other measured facts set the stage. First, the trained decay rate is ≈ 0.89 , not the belief-geometry rate $\zeta = 0.55$, and replacing the entire trained pattern by the ideal ζ -geometric one changes eval CE by $+0.00002$: near the optimum the loss barely cares about the pattern’s shape, which is exactly why training is free to leave gauge-flavored structure in the QK weights. Second, its TopK SAE allocates 24 latents to positions and only 2 to the three tokens (three centered one-hots span a plane) — and token-involving term-pairs carry just 12.3% of the score variance here, against 57–75% in the mixture toy. Same task family, same training recipe, an attribution share differing by a factor of five: shares are properties of the run, not of the computation.

Then the dial itself (`code/phase_a/gauge_dial.py`): apply the re-split $e(z) \rightarrow e(z) + tu$, $p \rightarrow p - tu$ with $u = -\bar{e}$ and sweep t from -1 to 2 . At every dial value the model’s logits are identical to float precision ($\leq 4.2 \cdot 10^{-7}$). Meanwhile the pedestal — the coefficient every key-side cut inherits — moves linearly from $+0.196$ through zero (at $t \approx 1.2$) to -0.070 . And the *same named intervention*, “cut token feature 0 on the key side at gain 1,” follows it: ΔCE falls from $+9.6 \cdot 10^{-4}$ to $+3 \cdot 10^{-5}$ at the canonical gauge and rises again past the crossing — a $32\times$ swing in the causal effect of one fixed operation on one fixed function, V-shaped in $|\beta_{\text{ped}}|$, with the pattern disturbance (TV $0.126 \rightarrow 0.032 \rightarrow 0.062$) tracing the same V. The floor at the bottom of the V is the $O(c\varepsilon)$ residue of the 9% leak, and the pedestal’s spread across the token features stays in $0.015\text{--}0.063$, small against the dialed pedestal everywhere except right at the crossing (where the mean passes through zero and any spread exceeds it): universality holds as well as the leak allows. Figure 1 is this paragraph as a picture.

The gauge dial: identical model, dialed FRA-QK — cut effects measure gauge, not computation (1-layer Mess3, idealized basis)

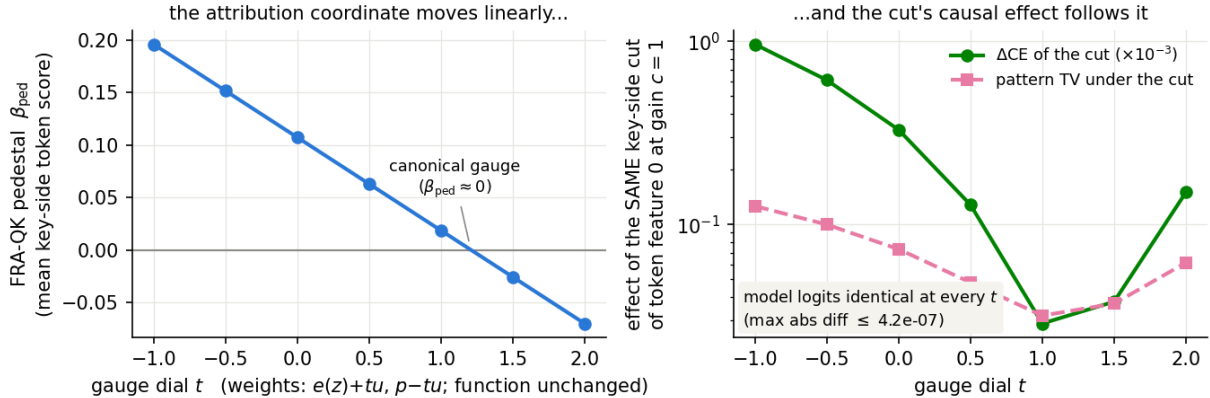


Figure 1: The gauge dial on the trained 1-layer Mess3 model. Left: the embedding re-split moves the FRA-QK pedestal linearly through zero while the model’s function is bit-identical at every dial value. Right: the behavioral effect of the *same* key-side FRA-QK cut follows the dialed coordinate, collapsing at the canonical gauge.

3.4 Where FRA-QK genuinely works: the four-context parallelogram

The gauge theorem is conditional on the optimal pattern being token-independent, and that is a property of the *data*. On induction data the condition fails, provably, and FRA-QK acquires a real handle — this is the regime where FRA interventions beat steering baselines in earlier experiments (`fra_win`: induction, in-context backdoors). The proof that lag-only patterns cannot do induction is a four-line argument worth internalizing.

The induction task: predict the successor of the most recent earlier occurrence of the current token. Fix two context positions $s_1 < s_2$, a query token A at the end, and fixed successor tokens X (after s_1) and Y (after s_2). Consider four contexts differing only in the tokens at (s_1, s_2) :

context	tokens at (s_1, s_2)	task demands	margin for X over Y
C_1	(A, B)	most recent A is at $s_1 \Rightarrow$ predict X	$g_1 \geq m$
C_2	(B, A)	most recent A is at $s_2 \Rightarrow$ predict Y	$g_2 \leq -m$
C_3	(A, A)	most recent A is at $s_2 \Rightarrow$ predict Y	$g_3 \leq -m$
C_4	(B, B)	A appears nowhere: no preference	forced: $g_4 \geq m$

For any lag-only pattern with per-key values, the head output is $a(d - s_1) \nu(\text{token at } s_1) + a(d - s_2) \nu(\text{token at } s_2) +$ (terms common to all four). The position-wise token multisets of $\{C_1, C_2\}$ and $\{C_3, C_4\}$ coincide — each has one A and one B at s_1 , one A and one B at s_2 — so the outputs obey the parallelogram identity $\mathbf{c}^{(1)} + \mathbf{c}^{(2)} = \mathbf{c}^{(3)} + \mathbf{c}^{(4)}$ exactly, and so do the X-versus-Y logit margins: $g_1 + g_2 = g_3 + g_4$. The task’s first three demands then force $g_4 = g_1 + g_2 - g_3 \geq m$: the model must confidently predict X in the one context that contains no A at all, where the truth prefers neither. Every lag-only model pays a fixed cross-entropy penalty; the optimum must gate its scores on content ($\eta \mathbb{I}[z_s = z_d]$ does the job); the feature $\times$ feature mass acquires a loss-pinned, gauge-*invariant* component; and cutting it moves behavior by $O(1)$.

So the boundary of FRA-QK’s usefulness is a crisp data property: **does the task force the pattern to depend on content?** Induction: yes. Belief updating over a stationary HMM — and, a fortiori, counting block tags: no. On the counting side there is a second, independent protection mechanism, which is the next section.

4 Puzzle 1, part two: running averages protect themselves

The gauge theorem says FRA-QK cut coefficients are unpinned; it leaves open the possibility that a large-gain edit, whatever its provenance, might still damage the concept by brute pattern distortion. The experiment says no: at the interface where keys carry raw token tags, every FRA-QK cut at every gain leaves $\text{RF} \leq 0.10$ while the *grammar* collateral CF climbs to 0.30. The concept is specifically protected, and the protection has a one-line mechanism.

The concept content of a key is its running average — and running averages barely differ between keys. Recall Eq. (1): the posterior mean after s tokens is $(\alpha + n(s))/(10 + s + 1)$. Between key s and query d it can drift by at most

$$|\hat{\omega}_k(d) - \hat{\omega}_k(s)| \leq \frac{d - s}{\kappa + d} \quad (\kappa = 10 \text{ in the toy}), \quad (5)$$

deterministically — each new token moves a count-based average by at most one part in $\kappa + d$ — and the typical (mean-square) drift is smaller still, of order $\sqrt{(d - s)/((\kappa + s)(\kappa + d))}$, by the martingale property of posterior means. You can see the drift shrinking directly in the worked sequence: $\hat{\omega}_1 = 0.611$ at $t = 16$, 0.78 at $t = 64$, 0.83 at $t = 128$ — adjacent stretches of keys agree about the concept to a few percent, and ever more tightly as the sequence grows. Now replay the slogan of Section 2: a score edit moves a readout only through the covariance of the edit with that readout’s per-key values. For a concept readout, the per-key values are (affine in) the running averages $\hat{\omega}(s)$, which agree across the attended window up to the drift bound. *Any* pattern edit, at *any* gain, only reshuffles weight among keys that nearly agree — approximate Invariance 2. Quantitatively, for any edit δ at gain c with $E = c \text{ osc}(\delta)$,

$$|\Delta(\text{concept readout})| \leq (e^E - 1) \kappa_\omega \frac{L_d}{\kappa + d}, \quad L_d = \text{the pattern’s mean lag}, \quad (6)$$

so a geometric attention window (L_d bounded) gives $O(1/d)$ protection, and even a flat counting window gives $O(1/\sqrt{d})$ typical protection. Grammar readouts enjoy no such shield: their per-key values are functions of the token identity itself, $\psi(z_s)$, which jumps at full spread Γ between adjacent keys — block tags flip between 0, 1, 2 constantly — so pattern edits move grammar by $\Theta(\Gamma)$ with no decay in d .

The predicted signature is an asymmetry, and the toy shows exactly it:

- Concept protected, grammar exposed: $\text{RF} \leq 0.10$ across the entire FRA-QK grid, while CF reaches 0.30. The cut damages the machinery it can reach (belief tracking within components) and misses the machinery it was aimed at.
- Protection tightens with position, exposure does not: for the key-side `omega_top4` cut at $c = 1$, per-position concept damage falls $0.0075 \rightarrow 0.0018 \rightarrow 0.0006$ from early to mid to late positions (at $c = 2$: $0.028 \rightarrow 0.010 \rightarrow 0.0014$) — a 13–20 $\times$ early-to-late drop — while CF stays order-one throughout.
- The decay follows the predicted power law on the primary run, and fails to on the replication. The clean pattern’s mean lag is bounded ($7.5 \rightarrow 10.5$): a geometric-type window, for which the theorem predicts $\text{MSE damage} \propto (\kappa + t)^{-2}$; the measured exponent at $c = 2$ is -1.97 (95% CI $[-2.12, -1.79]$). On the independently

trained `seed43` run, damage instead *rises* with position at $c = 2$ (exponent $+0.42$) — consistent with the bound (whose $(e^E - 1)$ prefactor can grow with position, and which covers only the direct head-output channel in this 3-layer model) but a real limit on the simple exponent story: one run of two.

FRA-QK on the aggregation toy: the softmax response calculus, verified (phase2_L0, key-side omega_top4 cut)

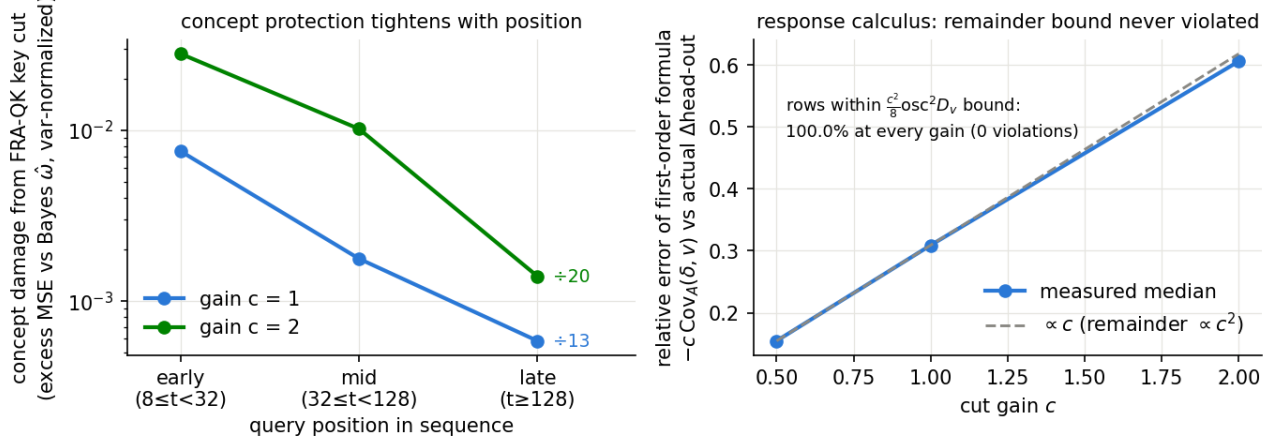


Figure 2: The QK side, verified on the mixture toy. Left: concept damage from the key-side FRA-QK cut falls 13–20 $\times$ from early to late positions — aggregation protection tightening exactly as the drift bound prescribes. Right: the first-order covariance formula’s error grows linearly in c (remainder $\propto c^2$) and the $\frac{c^2}{8} \text{osc}^2 D_v$ bound is never violated on any row.

Putting the two parts of Puzzle 1 together: the feature $\times$ feature attribution was gauge (Section 3), so cutting it had no reason to work; and even a cut with real oscillation lands on a readout whose keys agree (this section), so it had no way to work. Counting is doubly safe from routing surgery: *the concept is carried by what is summed, never by who attends to whom.*

5 Puzzle 2: the OV “null” is signal attenuation

5.1 One number governs the whole gain sweep

The FRA-OV cut subtracts c times a *fixed* vector field — the S -content transported by this layer, computed once from the clean run — so the intervened activation is exactly affine in c by construction. Suppose for a moment that everything downstream of the hookpoint responds linearly too. Then the model’s concept signal $m(t)$ (block-mass minus prior) can only be scaled:

$$m_c(t) = (1 - c\rho) m(t), \quad (7)$$

where ρ is the *path share*: the fraction of the total concept flow into the output that passes through the cut path. Everything in the scoreboard’s OV rows follows from this one line:

- **Exact severing under-reaches.** At $c = 1$ the signal shrinks to $(1 - \rho)$ of itself. The toy’s cut path carries a minority share, so severing it exactly leaves most of the concept flowing through parallel routes (the direct residual stream, other latents, later-layer re-aggregation): RF 0.06.
- **The null is a calibrated cancellation.** The signal vanishes at $c^* = 1/\rho$: past $c = 1$ the edit injects an *anti-concept* signal down one path, tuned to cancel the concept arriving by all the others. With the sweep-fit effective share $\rho \approx 0.25$, $c^* \approx 4$ — the observed null.
- **The metrics inherit the scaling.** Tracking R^2 falls as $R_{\text{clean}}^2 - (c\rho)^2$, and $\text{RF} \approx (c\rho)^2$ because cross-entropy is locally quadratic in signal scaling around calibration. Sanity check at $c = 1$: $(0.25)^2 = 0.0625$ against observed RF 0.062.
- **Presence is untouched, necessarily.** Scaling the *expressed* signal down one path changes nothing about the concept content sitting in the residual stream and the parallel paths; a probe that refits simply reads it

there: $R^2 = 0.985$ at the null. And overdriving past c^* turns the edit into an anti-concept steering vector — at $c = 6$ the sequence’s true dominant component is pushed *below* its prior.

“Remove the concept” has quietly become two different tasks. The gain-tuned null removes the concept’s *use* — its expression in behavior — by destructive interference. The concept’s *presence* — its decodability from the activations — is a different quantity, which no pattern or path edit touches.

5.2 Predicting the sweep before running it

The attenuation law is falsifiable in a strong form: ρ is measurable from the *clean model alone*, by a single finite difference — inject an infinitesimal multiple of the cut direction, read the slope of the concept signal (the measurement moves by at most 10^{-4} across two orders of magnitude of the probe step). On checkpoint `phase2.L0` this gives

$$\hat{\rho} = 0.1745 \quad (\text{independent run } \texttt{seed43}: \hat{\rho} = 0.1320),$$

and carrying this single linearization to its exact second-order consequences (the linear response also has a component orthogonal to the concept signal — collateral that damages tracking without removing use — plus a clean-residual cross term; both computable ex ante, still with no intervention data) predicts the whole sweep. Against the observed one:

gain c	RF predicted	RF observed	track R^2 predicted	track R^2 observed
0.5	0.019	0.019	+0.977	+0.977
1.0	0.061	0.062	+0.943	+0.941
1.5	0.126	0.126	+0.889	+0.881
2.0	0.214	0.215	+0.815	+0.795

Through $c \leq 2$ the prediction is essentially exact — third-decimal agreement in RF — from one derivative of the clean model. The predicted null is $c^* = 4.91$ against the observed 4.00 (`seed43`: 5.90 vs 4.59). The $\sim 20\%$ gap is itself measured rather than mysterious: the downstream stack (an MLP, another attention layer, the softmax) bends along the injection ray at large c , and the effective secant share grows from 0.18 at $c = 1$ to 0.23 at $c = 6$ (31% relative nonlinearity at $c = 4$) — precisely the linearity assumption failing by a fifth, where the theory says the failure must show up.

The linearization even predicts the null’s fine structure before looking. The path share rises with position ($\hat{\rho} = 0.132$ early, 0.171 mid, 0.182 late), so a single aggregate gain must under-cancel early positions and over-cancel late ones — and the observed null indeed balances a +0.11 residual tracking early against -0.09 anti-tracking late. The “null” is an average of signed miscalibrations, and drifts off null on inputs whose signal strength differs from the calibration set (near-pure sequences get pushed anti-concept). A calibrated equilibrium, with the calibration visible.

The one-layer Phase A model shows the same structure with the belief geometry exposed. Its per-token OV vectors align with the predicted belief displacements $\mathbf{B}(\pi \mathbf{T}^{|z} - \pi)$ at $\cos = 0.91\text{--}0.93$ (position-stable to $\cos > 0.998$), its attention-output probe prefers the constrained update over full Bayes ($R^2 = 0.818$ vs 0.766), and the SAE-basis FRA decomposition of scores and attention output is exact to $2 \cdot 10^{-6}$ — the attribution objects are the theory’s objects. The interventions then reproduce the asymmetry at matched features: FRA-QK cuts of the two token latents at $c = 1$ move eval CE by at most +0.0006 (5% of the 0.012-nat information window, growing superlinearly, $\approx 2.5\times$ per gain doubling), while FRA-OV severing of the *same latents* at $c = 1$ costs +0.0092 — 77% of the window, a sixteenfold gap, with over-removal beyond severing ($c = 4$: $12.8\times$ the window). Content transported through the pattern is causally live; the pattern’s token-resolved bookkeeping is not.

5.3 The use/presence trichotomy

The three intervention families now line up as three different operations on three different objects:

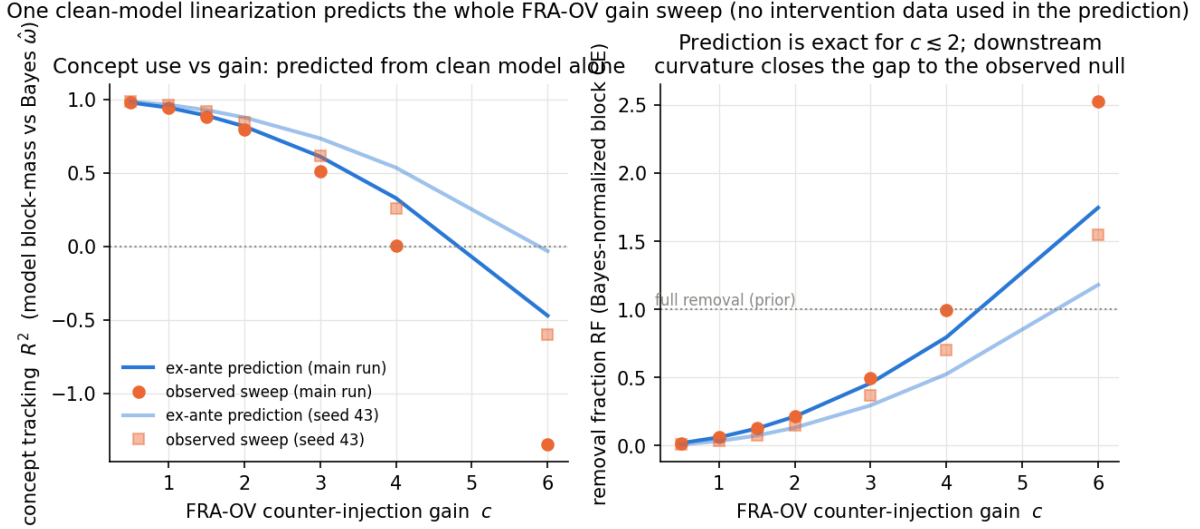


Figure 3: Ex-ante attenuation predictions (from one finite difference on the clean model; no intervention data) against the observed FRA-OV gain sweep, for both independently trained runs. The refined-linear curves track the observations essentially exactly through $c \leq 2$; the growing gap beyond is measured downstream curvature.

family	what it does to an aggregated concept	evidence
pattern edits (FRA-QK, any set, side, gain)	reshuffle weight among keys that agree; use protected by drift, presence untouched; grammar pays $\Theta(\Gamma)$	RF ≤ 0.10 , CF up to 0.30, presence 0.99
path-content edits (FRA-OV at gain c)	scale the expressed signal by $(1 - c\rho)$ along one path; use nulled at $c^* = 1/\rho$, presence untouched	RF 0.99 at $c \approx 4$, presence 0.985
content edits (SAE cuts at the hookpoint)	remove the aggregate content itself where it fires: the only family that reduces presence	RF 0.86, presence $0.99 \rightarrow 0.49$

Even the third family has a wall. The concept here is an aggregate statistic *of the content the grammar needs*: $\hat{\omega}$ counts block tags, and belief updating must route each symbol’s history to the right component using those same tags. A particle-filter computation of the ideal frontier gives the toy’s *entanglement tax*: an observer carrying zero ω -information achieves within-block CE 0.845 at best, against 0.7785 with full information — an irreducible 20.8% of the grammar range that *any* method must pay for full presence removal. Behavioral evals certify none of this: the $c \approx 4$ model passes every on-distribution behavioral test of concept use while carrying the concept at $R^2 = 0.995$. The audit that catches it is a presence probe, and the operative lesson of the whole toy is: **audit presence, not behavior**.

6 One dial between the two regimes

Sections 3–5 describe two seemingly different worlds: an *association* world (induction: content-gated patterns, FRA-QK has an $O(1)$ handle) and an *aggregation* world (mixture counting: flat token-independent patterns, FRA-QK inert, FRA-OV = attenuation). One dial connects them: the spectral gap of the data process.

A hidden chain’s transition operator has eigenvalues; the stationary eigenvalue 1 carries what never decays, and eigenvalues $\lambda < 1$ carry evidence with memory horizon $\sim 1/(1 - \lambda)$. The mixture concept is precisely extra eigenvalue-1 structure: the per-sequence label ω never changes, so the joint hidden process (mixture label $\otimes$ within-component state) has eigenvalue 1 with multiplicity 3, and the concept subspace *is* that eigenspace. The grammar lives in the $\lambda < 1$ modes. Aggregation is the $\lambda \rightarrow 1$ corner of the same framework, and the optimal attention profile interpolates continuously: solving the one-layer theory for a mode at eigenvalue λ gives an optimal geometric window with rate

$\eta(\lambda)$ obeying, as $\lambda \rightarrow 1$,

$$1 - \eta \simeq \sqrt{\frac{2m}{\varphi}} \sqrt{1 - \lambda} \quad \Rightarrow \quad \text{window } \frac{1}{1 - \eta} \propto \frac{1}{\sqrt{1 - \lambda}}. \quad (8)$$

The closed form matches direct numerical optimization to six decimal places across $\lambda \in [0.5, 0.9999]$, a free 300-lag profile optimization recovers the geometric shape exactly, and the window grows from 1.4 to 75 over that range (Figure 4). In the strict limit the optimal profile is exactly flat: counting, recovered as a spectral limit.

Why aggregation concepts get flat, content-blind attention (P3 two-state closed form, verified numerically)

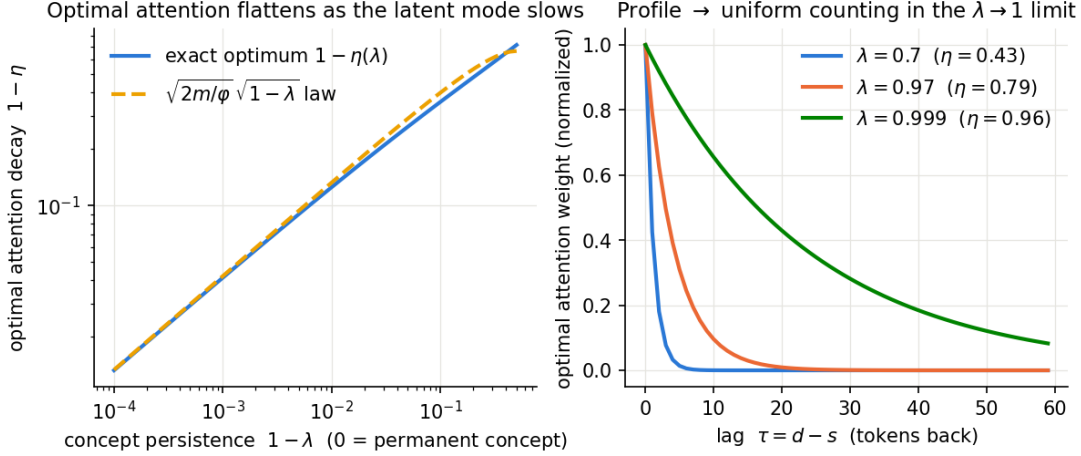


Figure 4: The flattening law. As the data eigenvalue $\lambda \rightarrow 1$, the optimal attention rate η obeys $1 - \eta \propto \sqrt{1 - \lambda}$ and the effective window $1/(1 - \eta)$ diverges: sharp associative patterns flatten into counting. Closed form against numerical optimization, agreement to six decimals; window $1.4 \rightarrow 75$ across $\lambda \in [0.5, 0.9999]$.

Reading the dial from left to right gives the field guide:

- **λ well below 1 (association regime).** Short memory, sharp patterns; tasks like induction force content gating (Section 3.4); the pattern is a causal carrier and FRA-QK edits it usefully; feature $\times$ feature attribution has a loss-pinned component.
- **$\lambda \rightarrow 1$ (aggregation regime).** Long flat counting windows; the pattern is token-independent, so FRA-QK coefficients are gauge (Section 3); the concept rides in aggregated content whose keys agree, so all pattern edits are drift-bounded (Section 4); FRA-OV acts as linear attenuation with a path-share null (Section 5); and only content edits touch presence, up to the entanglement tax.

Every broad, slowly-varying in-context concept — a topic, a persona, a per-document property estimated by accumulation — is an aggregation-regime object. The toy’s prediction for such concepts in real models is the scoreboard of Section 1: spectral and attribution diagnostics will find mass; pattern surgery will find no handle; path severing will under-reach; gain-tuned cancellation will fake removal; and only content-level edits will move what a retrained probe sees.

7 Honest limits

Four items bound how far the story above is proved, as opposed to measured.

1. **The null gain is predicted to $\sim 20\%$, and the residue is measured curvature.** The ex-ante linearization matches the sweep to the third decimal through $c \leq 2$, but c^* sits at 4–6, where the downstream map bends: the secant path share grows $0.18 \rightarrow 0.23$ over $c \in [1, 6]$. An exact ex-ante c^* needs more than one linearization of the downstream stack; we report the one-derivative theory with its error measured, on both runs.
2. **One idealized-basis prediction fails on the trained SAE, for a reason the theory names.** In the idealized basis, cutting the union of *all* token features on the key side is a per-row constant — exactly inert

(Invariance 1). On the trained TopK SAE, the all-latents key cut is *active*: its per-row score oscillation per unit of subtracted mass is $1.48\times$ that of the small `omega_top4` set (the idealized theorem predicts a ratio far below 1), and it does more at matched gain (RF 0.020, CF 0.231 at $c = 1$, versus 0.0023/0.045). The mechanism: trained latents at this interface carry positional and aggregate content — running counts correlate with position — so the latent union subtracts part of the attention *profile* itself, which no token-feature union does in the idealized basis. Per-feature statements (the covariance formula, the position scaling) transfer to the SAE basis; set-union statements are basis-sensitive.

3. **Lag-only optimality for softmax/CE is a conjecture.** It is proved for the free-profile MSE surrogate, and trained softmax models measure as token-independent to precision — but the CE case is open, and a model sitting slightly off the lag-only manifold has a small genuine content coupling, covered only by a measurable leak parameter. All gauge statements are exact *given* token-independence.
4. **Probe identifications are heuristic.** Reading the model’s concept channel as a scaled posterior mean, and key values as functions of the block tag, is supported by probe R^2 rather than derived. The *rates* in d (the drift bounds, the position scaling) are independent of these identifications; the constants inherit them.

Two habits the toy should install regardless: sweep intervention gains past $c = 1$ before declaring a null result (the operating point that mattered here sat outside the pre-registered grid); and when a behavioral null appears, retrain a probe before believing anything was removed.